1125 labmeeting - 미공개

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1 previous

2 model description

2.1 data likelihood

$$\begin{cases} \log W_{ij} | i_{ij} = 1 \sim \mathcal{N}(\eta_{ij}, \sigma_0^2) \\ \log W_{ij} | i_{ij} = 0 \sim \mathcal{N}(\eta_{shrink}, \sigma_{shrink}^2) \end{cases}$$

such that, comment: 이 구조에서 $\mathcal{N}(\eta_{shrink}, \sigma_{shrink}^2)$ distribution 이 위의 η_{ij} 기반 distribution 이 설명 잘 못하는 것을 효과적으로 구분하지 못할 것 같다. noise 가 normal distribution 을 가진다는 가정 자체가 좀 unrealistic 하다. 해당 구조의 목적은 $\mathcal{N}(\eta_{ij}, \sigma_0^2)$ 으로 잘 설명하지 못하는 - 혹은 전체 구조를 망가뜨리는 edge 구조를 다른 distribution 으로 보내서 설명하게끔 하려는 것인데, 그렇다면 normal 보다는 조금 더 유연한 구조가 필요해 보인다.

$$\eta_{ij} = \theta_i + \theta_j - \beta d(z_i, z_j)$$

$$W_{ij} \in (0, \infty)$$

2.2 prior distribution

and prior distribution is,

- $\eta_{shrink} \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$
- $\sigma_{shrink}^2 \sim \mathcal{IG}(a = 0.05, b = 0.05)$

- $\sigma_0^2 \sim \mathcal{IG}(a_0 = 0.05, b_0 = 0.05)$
- $\theta_i \sim \mathcal{N}(0, 1)$
- $\beta \sim \text{Gamma}(a_1, b_1)$
- $z_i \sim N(0, I\sigma_1^2)$
- $\sigma_1^2 \sim \mathcal{IG}(a_1 = 0.05, b_1 = 0.05)$
- $i_{ij} \sim \text{Bernoulli}(P)$
- $P \sim \text{Beta}(a_2 = 1, b_2 = 1)$

and likelihood depends on indicator variable i_{ij} and suppose $Y_{ij} = \log W_{ij}$ such that W_{ij} is non-negative edge weight of target network.

2.3 MCMC sampling steps

2.3.1 update η_{shrink}

let $\mathcal{I}$ is set of all possible i_{ij} and $\mathcal{I}_0$ is set of all $i_{ij} = 0$ and $\mathcal{I}_1$ is set of all $i_{ij} = 1$.

$$P(Y|\eta_{shrink}, \mathcal{I} = \mathcal{I}_0)P(\eta_{shrink}) \propto \ell(Y|\eta_{shrink}, \mathcal{I} = \mathcal{I}_0) + \log P(\eta_{shrink})$$

therefore, with $A = \left(\frac{\sum_{(i,j):i_{ij}=0} 1}{\sigma^2_{shrink}} + 1 \right), B = \left(\frac{\sum_{(i,j):i_{ij}=0} Y_{(i,j)}}{\sigma^2_{shrink}} \right)$,

$$\eta_{shrink}| \cdot \sim \mathcal{N}\left(\frac{B}{A}, \frac{1}{A}\right)$$

2.3.2 update σ^2_{shrink}

with same setting of i_{ij} set,

$$P(\sigma_{shrink}|\cdot) \propto L(Y|\mathcal{I} = \mathcal{I}_0, \cdot)P(\sigma^2_{shrink})$$

and since $\sigma^2_{shrink} \sim \mathcal{IG}(a, b)$,

$$\sigma^2_{shrink}| \cdot \sim \mathcal{IG}\left(\frac{\sum_{(i,j):i_{ij}=0} 1}{2} + a, \frac{1}{2} \sum_{(i,j):i_{ij}=0} (y_{(i,j)} - \eta_{shrink})^2 + b \right)$$

2.3.3 update σ_0

with same setting of i_{ij} set,

$$P(\sigma_0^2|\cdot) \propto L(Y|\mathcal{I} = \mathcal{I}_1, \cdot)P(\sigma_0^2)$$

and since $\sigma_0^2 \sim \mathcal{IG}(a_0, b_0)$,

$$\sigma_0^2|\cdot \sim \mathcal{IG}\left(\frac{\sum_{(i,j):i_{ij}=1} 1}{2} + a_0, \frac{1}{2}\sum_{(i,j):i_{ij}=1}(y_{(i,j)} - \eta_{(i,j)})^2 + b_0\right)$$

2.3.4 update θ_i

with same setting of i_{ij} set, suppose that ϵ_i is set of edges that comes from node i and $i_{ij} = 1$.

$$\begin{aligned} P(\theta_i|\cdot) &\propto L(Y|\cdot)P(\theta_i) \propto \ell(Y|\cdot) + \log P(\theta) && (1) \\ &= \sum_{(i,j)\in\epsilon_i} -\frac{1}{2\sigma_0^2}(y_{(i,j)} - \eta_{(i,j)})^2 - \frac{1}{2}\theta_i^2 && (2) \\ &= \sum_{(i,j)\in\epsilon_i} -\frac{1}{2\sigma_0^2}(y_{(i,j)} - (\theta_j - \beta d(z_i, z_j)) - \theta_i)^2 - \frac{1}{2}\theta_i^2 && (3) \end{aligned}$$

therefore, with $A = \left(\frac{\sum_{(i,j):i_{ij}\in\epsilon_i} 1}{\sigma_0^2} + 1\right), B = \left(\frac{\sum_{(i,j):i_{ij}\in\epsilon_i}[y_{(i,j)} - (\theta_j - \beta d(z_i, z_j))]}{\sigma_0^2}\right)$

$$\theta_i|\cdot \sim \mathcal{N}\left(\frac{B}{A}, \frac{1}{A}\right)$$

2.3.5 update β - MH

$$\text{MH ratio of } \beta = \frac{\ell(Y|\beta', \mathcal{I} = \mathcal{I}_1, \cdot) + \log P(\beta') + \log\beta}{\ell(Y|\beta, \mathcal{I} = \mathcal{I}_1, \cdot) + \log P(\beta) + \log\beta'}$$

with proposal distribution such that $W = \log\beta, \quad W \sim \mathcal{N}(\log\beta, \text{prop\_beta})$ and β' is proposed beta variable.

2.3.6 update z_i - MH

$$\text{MH ratio of } z_i = \frac{\ell(Y|z_i', \mathcal{I} = \mathcal{I}_1, \cdot) + \log P(z_i')}{\ell(Y|z_i, \mathcal{I} = \mathcal{I}_1, \cdot) + \log P(z_i)}$$

using proposal distribution such that $z_i \sim \mathcal{N}(z_i, \text{prop\_z})$

2.3.7 update i_{ij}

$$P(i_{ij}|Y,\cdot) = \frac{P(y_{ij}|i_{ij}=1,\cdot)P(i_{ij}=1)}{\sum_{k=0}^{1} P(y_{ij}|i_{ij}=k,\cdot)P(i_{ij}=k)}$$

with $P(i_{ij}=1) = p, \quad P(i_{ij}=0) = 1-p$

comment: (광희님) i 가 update 될 때 너무 network 구조를 못 받고 있는 것 같음. 지금은 i 를 기존 η_{ij} 기반 구조가 η_{shrink} 기반 구조보다 Y_{ij} 를 얼마나 잘 설명하는가만을 보고 있는데, 이렇게 구분하게 되면 '비슷하게 설명되는' edge 에 대해서 명확하게 구분 안 될 수도 있고, 이 문제가 전체 MCMC chain 수렴을 방해할 수 있다. 그래서 최대한 잘 구분할 수 있게 해야 할 것 같은데, 이 과정에서 network 구조에서 얻을 수 있는 정보를 더 반영하면 좋을 것 같다.

2.3.8 update p

with $p \sim \text{Beta}(\alpha,\beta)$, $i_{ij} \sim \text{Bernoulli}(p)$ therefore,

$$P(p|\cdot) \propto p^{\sum i_{ij}+\alpha-1} \times (1-p)^{\sum (i_{ij}-1)+\beta-1}$$

so,

$$p|\cdot \sim \text{Beta}\left(\sum_{(i,j)\in\mathcal{I}} i_{ij}+\alpha,\ \sum_{(i,j)\in\mathcal{I}} (1-i_{ij})+\beta\right)$$

References